City Brain Convergence — Sensors & Actuators Data Dictionary

Guardian layer · shared-contract reference · Triplet working draft

This document catalogs every sensor and actuator type the Digital Twin is expected (or likely) to expose, with a canonical data format for each. It covers the five sensor types and five actuator types named explicitly in the challenge brief (**Required**), plus a broader, predicted set covering the rest of what a real smart city would run (**Extended**) — so Guardian's detection logic, and the shared contract itself, aren't caught flat-footed if the Twin team implements more than the minimum, or if Phase 2's organizer-run scenarios introduce something outside the original five.

All readings and calls use the common envelope defined in Section 1 regardless of type. Field names, units, and zone granularity below are a proposal for triplet negotiation, not a settled standard — see Section 8.

1. Common envelope

Every sensor reading, regardless of type, carries this envelope. Type-specific fields live inside `value`.

Field	Type	Example	Notes
`contract_version`	string (date)	`"2026-09-17"`	Versioned from day one; a mismatch is self-documenting
`sensor_id`	string	`"wl-port-03"`	Stable unique ID per physical/simulated device
`type`	enum	`"water_level"`	One of the sensor types in Sections 2–3
`zone_id`	string	`"zone-07"`	City zone the sensor belongs to (see Section 7)
`timestamp`	ISO-8601 string	`"2026-11-02T14:03:21Z"`	UTC, millisecond precision
`sequence_no`	integer	`4821`	Monotonic per sensor; used for replay detection

Field	Type	Example	Notes
`value`	object	*(type-specific, see tables below)*	The actual reading
`provenance`	object	`{"signed_by":"twin","hmac":"..."}`	HMAC over the payload, signed by Twin

Every actuator call carries this envelope:

Field	Type	Example	Notes
`contract_version`	string (date)	`"2026-09-17"`	Same versioning scheme as readings
`actuator_id`	string	`"valve-port-12"`	Stable unique ID per actuator
`action`	enum	`"set_valve_state"`	One of the actions in Sections 4–5
`params`	object	*(action-specific)*	Arguments for the action
`requesting_layer`	enum	`"city_brain"`	`city_brain` or `guardian`
`justification`	string	`"flood threshold exceeded, zone-07"`	Human-readable reason, written to the audit log
`trust_ref`	string | null	`"verdict-88213"`	The Guardian verdict this call relied on; null only for Guardian's own direct containment calls

2. Sensors — required (per challenge spec)

2.1 Traffic camera

Field	Type	Range / unit	Example
`congestion_score`	float	0–1	`0.73`
`vehicle_count`	integer	count / interval	`42`
`avg_speed_kmh`	float	km/h	`18.5`

Field	Type	Range / unit	Example
`smoke_detected`	boolean	—	`false`

2.2 Water-level sensor

Field	Type	Range / unit	Example
`level_m`	float	metres	`0.42`
`flow_rate_m3s`	float (optional)	m³/s	`1.2`

2.3 Power-load meter

Field	Type	Range / unit	Example
`load_kw`	float	kW	`340.0`
`capacity_kw`	float	kW	`500.0`
`voltage_v`	float	V	`230.4`

2.4 Air-quality monitor

Field	Type	Range / unit	Example
`aqi`	integer	0–500	`58`
`pm25_ugm3`	float	µg/m³	`12.3`
`pm10_ugm3`	float	µg/m³	`21.0`

2.5 Emergency-call feed

Field	Type	Range / unit	Example
`call_id`	string	—	`"call-55021"`

Field	Type	Range / unit	Example
`category`	enum	fire | medical | police | flood | other	`"fire"`
`caller_confidence`	float (optional)	0–1	`0.6`

3. Sensors — extended (predicted, for completeness and resilience)

Category	Sensor	Key fields	Unit / range
Mobility	Inductive loop / speed sensor	`speed_kmh`, `vehicle_class`	km/h
Mobility	Parking occupancy sensor	`occupied` (bool), `spot_id`	—
Mobility	Public transit tracker	`route_id`, `occupancy_pct`, `delay_min`	% / min
Mobility	Pedestrian counter	`count_5min`	count
Utilities	Grid frequency/voltage sensor	`frequency_hz`, `voltage_v`	Hz / V
Utilities	Renewable generation sensor	`output_kw`, `capacity_kw`	kW
Utilities	Gas leak sensor	`concentration_ppm`, `threshold_breach` (bool)	ppm
Utilities	Smart building water meter	`consumption_l`	litres
Environmental	Rainfall sensor	`mm_per_hr`	mm/h
Environmental	Soil moisture sensor	`moisture_pct`	%
Environmental	Noise level sensor	`db_level`	dB
Environmental	Temperature/humidity sensor	`temp_c`, `humidity_pct`	°C / %
Environmental	UV index sensor	`uv_index`	0–11+

Category	Sensor	Key fields	Unit / range
Public safety	Fire/smoke detector (building)	`smoke_detected` (bool), `temp_c`	— / °C
Public safety	Structural health sensor (bridge/building)	`vibration_mms`, `tilt_deg`	mm/s / °
Public safety	CCTV anomaly flag	`anomaly_type`, `confidence`	enum / 0–1
Public safety	Gunshot detection sensor	`detected` (bool), `confidence`	— / 0–1
Waste & services	Smart bin fill-level sensor	`fill_pct`, `bin_id`	%
Waste & services	Streetlight status sensor	`brightness_pct`, `fault` (bool)	%
Waste & services	Manhole/sewer sensor	`blockage_detected` (bool), `level_cm`	— / cm

4. Actuators — required (per challenge spec)

4.1 Police / ambulance dispatch

`action: "dispatch_unit"` · `params: {unit_type: police|ambulance|fire, zone_id, incident_id}` Result: `{status: dispatched|rejected|duplicate, eta_minutes, unit_id}`

4.2 Traffic-signal control

`action: "set_signal_phase"` · `params: {intersection_id, phase: green|red|yellow|flashing, duration_s}` Result: `{status, reverts_at}`

4.3 Water-valve control

`action: "set_valve_state"` · `params: {valve_id, state: open|closed, flow_target_pct?}` Result: `{status, current_state}`

4.4 Power-grid switching

`action: "set_grid_connection"` · `params: {zone_id, state: connected|disconnected}` Result: `{status, affected_load_kw}`

4.5 Device isolation / quarantine

`action: "quarantine_device" | "isolate_sensor" | "clear_quarantine"` · `params: {device_id, reason}` Requesting layer is normally `guardian`; `trust_ref` may be null since Guardian can self-authorize containment. Result: `{status, quarantined_until | cleared}`

5. Actuators — extended (predicted)

Category	Actuator	`action`	Key params
Mobility	Variable message sign control	`set_sign_message`	`sign_id`, `message_text`
Mobility	Parking barrier/gate control	`set_barrier_state`	`barrier_id`, `state: open|closed`
Mobility	Public transit reroute/dispatch	`reroute_transit`	`route_id`, `new_path`
Utilities	Streetlight control	`set_streetlight`	`light_id`, `brightness_pct`, `on`
Utilities	EV charging station control	`set_charging_station`	`station_id`, `enabled`, `max_kw`
Utilities	Pump station control	`set_pump_state`	`pump_id`, `state: on|off`, `flow_target`
Water infrastructure	Floodgate control	`set_floodgate`	`gate_id`, `position_pct`
Waste & services	Waste collection dispatch	`dispatch_collection`	`bin_ids[]`, `route`
Security / comms	Public alert broadcast	`broadcast_alert`	`message`, `channel: sms|siren|app`, `zone_id`

Category	Actuator	`action`	Key params
Security / comms	Network access revocation	`set_network_access`	`client_id`, `action: revoke|restore`

6. Zone & topology model

Every sensor and actuator belongs to a `zone_id` (e.g. `zone-07`) rather than raw GPS coordinates, keeping the model simple enough to simulate while still letting City Brain and Guardian reason about locality — "is this reading's zone adjacent to an active incident's zone" is a cheap, useful check regardless of how many zones the Twin implements. The exact number and shape of zones (a grid, named districts, or an actual road-graph) is a Twin-side decision; the contract only requires that `zone_id` be a stable, resolvable string every layer can reference without hard-coding geometry.

7. Guardian validation mapping

How Guardian's standard checks (Section 6 of the Phase 0 report) apply across sensor categories, with starting-point thresholds to refine during testing:

Sensor category	Primary risk	Guardian check	Starting threshold
Traffic camera	Fake congestion spike	Rate-of-change + cross-check vs. speed sensor	`congestion_score` jump >0.5 in <5s
Water-level	Flood spoofing	Rate-of-change + cross-check vs. rainfall sensor	Rise >0.2m per 5s with no matching rainfall
Power-load	Load spoofing / DoS	Rate-of-change + capacity ceiling	`load_kw` > `capacity_kw` or jump >30% per tick
Air-quality	Fire spoofing	Cross-check vs. camera smoke flag	AQI spike with no `smoke_detected` correlation
Emergency-call feed	Fake call flood	Arrival-rate limiting + duplicate detection	>5 calls/zone/minute, or identical `call_id` reused

Sensor category	Primary risk	Guardian check	Starting threshold
All types	Replay / tampering	`sequence_no` monotonicity + provenance HMAC	Any non-increasing `sequence_no` or HMAC mismatch
All types	Device dropout	Heartbeat monitoring	No reading within 2× expected interval

8. Open items for triplet negotiation

- Exact zone/grid granularity and whether Twin models a road-graph or flat zones.
- Confirming field names and units above with the Twin and City Brain teams before freezing `contract_version`.
- Whether extended sensors/actuators (Sections 3, 5) are in scope for the Phase 1 demo or reserved as Phase 2 stretch additions.
- Location format for emergency calls — `zone_id` alone, or `zone_id` plus approximate coordinates, depending on how precise Twin's incident placement is.